

Question number	Answer	Additional guidance	Mark
8(a)	An explanation that includes the following points: • {alternative versions / alleles} of a gene (1) • due to changes in the base sequence of the DNA (1)	ALLOW sections of DNA ALLOW differences in exons / introns of a gene ALLOW change in nucleotide sequence of DNA	Expert (2)

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Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes four of the following points: • (more) mutations change the shape of (more) proteins (1) • when these are membrane proteins (1) • (more of these) proteins are recognised as antigens when presented on antigen presenting cells (1) • triggering an immune response (1) • that includes production of T killer cells (that will recognise and destroy more cancer cells) (1)	Accept non self proteins ALLOW description of an immune response ALLOW cytotoxic T cells IGNORE lymphocytes	Expert (4)

Question number	Answer	Additional guidance	Mark
8(c)	An explanation that includes two of the following points: - CT uses x-rays to produce a (low resolution) image (of soft tissue structures) (1) - the tumor is a soft tissue/ has a different density (from other tissues) (1)		Expert (2)

Question number	Answer	Additional guidance	Mark
8(d)	An answer that includes two of the following points: - cancer cells produce proteins that act as checkpoints (1) - checkpoint inhibitors stop the checkpoint proteins from blocking immune cell activation (1) - allowing the (tumor-infiltrating) lymphocytes to be activated to kill the tumor cells (1)	ALLOW checkpoint inhibitors allow activation of immune cells	Expert (2)

Question number	Answer	Additional guidance	Mark
8(e)	An explanation that includes the following points: - the gene variant has an altered DNA base sequence (1) - produces a {polypeptide/protein} with a different primary structure (1) - enzyme produced will be folded differently (1) - therefore the enzyme's active site will no longer fit clopidogrel (1)	ALLOW different 3D shape / arrangement/tertiary structure ALLOW substrate if in context of clopidogrel ALLOW fewer enzyme – substrate complexes formed	Expert (4)

Question number	Answer	Additional guidance	Mark
8(f)	An explanation that includes three of the following points: - capillaries allow for mass transport (1) - to overcome limitations of diffusion (1) - ensure (sufficient) {nutrients / oxygen} delivered to spinal cord tissue / stem cells (1) - named function of spinal cord tissue (1)	ALLOW correctly named nutrient in context ALLOW spinal cord cells / tissues need to respire	Expert (3)

Question number	Answer	Additional guidance	Mark
8(g)	An answer that includes three of the following points: • {growth factors / proteins} bind to receptors (on pluripotent stem cells) (1) • activating transcription factors (1) • causing the stem cell to proliferate/divide (1) • and differentiate (to carry out a particular function) (1)	ALLOW proteins act as transcription factors	Expert (3)

